Freysoldt's Review

A Lattice Model of Point Defects in the Grand Canonical Ensemble

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1 Aim and Main Result

Aim of this note. Freysoldt *et al.* [1] reviewed first-principles calculations for point defects in solids. The review did not derive the equilibrium defect concentration from a partition function, and named the minimized potential a free energy F^{conf} and a Gibbs energy G, although both potentials contain the reservoir terms of a grand potential. This note builds a lattice model of point defects, evaluates the grand canonical partition function of the model from the definition, and derives the grand potential and all the conjugate variables. Every modelling assumption is stated as an assumption, and every mathematical approximation has an error bound. Here, "the note" refers to the lecture note on the entropy maximization principle, and "the defect notes" refers to the earlier notes on the review in `freysoldt.pdf`. All symbols are those of the note and of the defect notes, and Table 3 at the end lists every symbol.

Main result. The grand potential of a crystal with N sites that can each hold one defect is

$$\Phi = \Phi_{\text{bulk}} - N k_B T \ln\Big(1 + g_X \, e^{-E^f/k_B T}\Big),$$

where Φ_{bulk} is the grand potential of the perfect crystal, E^f is the formation energy of one defect and g_X is the number of internal states of one defect. Sections 2 and 3 state the model and the ensemble. Section 4 derives this result, and Section 5 derives the conjugate variables. Sections 6 to 9 treat the approximations, the minimization route of the review, the fixed pressure and the charge states. Section 10 translates the symbols of the review.

2 The Lattice Model

System and reservoirs. The system is a crystal region with a fixed number N of lattice sites and the volume V (Fig. 1). The region exchanges energy with a heat bath at the temperature T, atoms of each species i with a reservoir at the chemical potential μ_i, and electrons with a reservoir at the electron chemical potential, the Fermi level E_F. These reservoirs are the reservoirs of Section 1 of the defect notes. The natural variables are therefore $(T, V, \{\mu_i\}, E_F)$, and N is a fixed parameter that sets the size of the region.

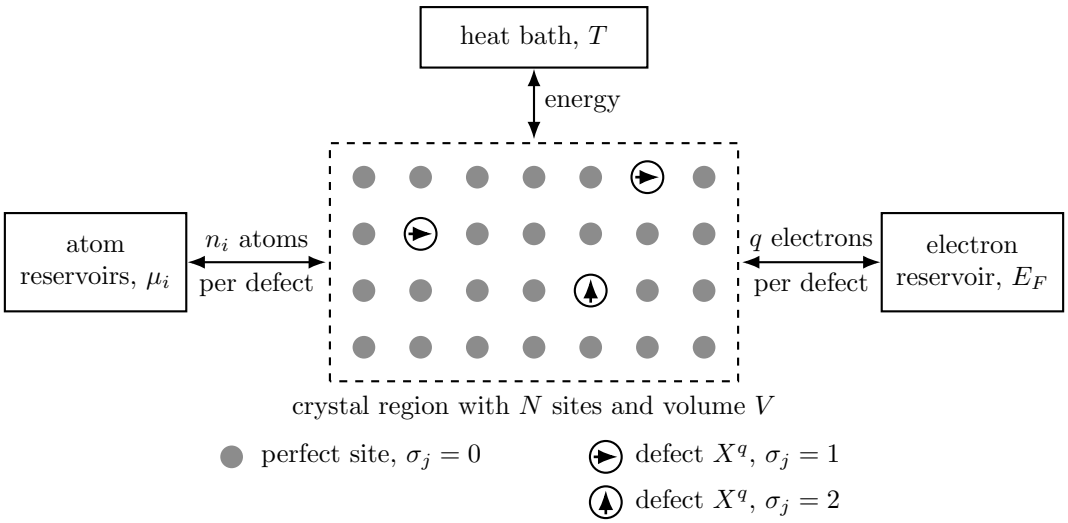


Figure 1: Lattice model of a crystal region with N sites in contact with a heat bath, atom reservoirs and an electron reservoir. Filled circles: perfect sites. Open circles with an arrow: defects X^q, with the arrow direction as the internal state for $g_X = 2$.

Defect configuration. The model contains one type of defect, the defect X in the charge state q, written X^q. Each site is either perfect or holds one defect. A defect at a given site has g_X internal states of equal energy, for example the symmetry-equivalent orientations of a split interstitial or the two spin directions of an unpaired electron. Here, σ_j denotes the state of site j,

$$\sigma_j = \begin{cases} 0, & \text{site } j \text{ is perfect,} \\ 1, \ldots, g_X, & \text{site } j \text{ holds a defect in the internal state } \sigma_j. \end{cases}$$

The defect configuration

$$\sigma = (\sigma_1, \sigma_2, \ldots, \sigma_N)$$

lists the states of all the sites, so the model has $(g_X + 1)^N$ configurations. Here, $N_X(\sigma)$ denotes the number of defects in the configuration σ,

$$N_X(\sigma) = \sum_{j=1}^{N} \big(1 - \delta_{\sigma_j,0}\big),$$

where $\delta_{\sigma_j,0}$ equals one for a perfect site and zero for a site with a defect.

Exchange with the reservoirs. The formation of one defect moves n_i atoms of species i from the reservoir into the crystal and releases q electrons into the electron reservoir. A negative n_i removes atoms, as for a vacancy. The crystal exchanges atoms and electrons with the reservoirs only by forming or removing defects. Here, N_i and N_e denote the numbers of atoms of species i and of electrons in the perfect crystal, and $N_{i,\sigma}$ and $N_{e,\sigma}$ denote the numbers in the configuration σ,

$$\boxed{N_{i,\sigma} = N_i + n_i\, N_X(\sigma), \qquad N_{e,\sigma} = N_e - q\, N_X(\sigma).} \tag{2.1}$$

The particle numbers therefore depend on the configuration only through the number of defects.

Microstates and the energy of a microstate. A configuration fixes where the defects are, but not the motion of the nuclei or the state of the electrons. With σ held fixed, the nuclei vibrate about the relaxed positions that belong to σ, and the crystal has a set of energy eigenstates. Here, ν labels these eigenstates, and the pair (σ, ν) labels a microstate in place of the index i of the note, because i labels the species here. The energy of the microstate (σ, ν) has three parts,

$$\boxed{U_{\sigma\nu} = E^{\mathrm{stat}}_{\sigma} + E^{\mathrm{vib}}_{\sigma\nu} + E^{\mathrm{el}}_{\sigma\nu}.} \tag{2.2}$$

- The static energy $E^{\mathrm{stat}}_{\sigma}$ is the energy of the crystal with the nuclei at rest at the relaxed positions and the electrons in the ground state. This energy is the total energy E_{tot} of a DFT calculation. The static energy contains the kinetic energy of the electrons and all the Coulomb interactions between the electrons and the nuclei, so the energy of the chemical bonds is part of the static energy and not a separate term.

- The vibrational energy $E^{\mathrm{vib}}_{\sigma\nu}$ is the energy of the nuclear motion. In the harmonic approximation,

  $$E^{\mathrm{vib}}_{\sigma\nu} = \sum_{k} \hbar\omega_{\sigma,k}\Big(\nu_k + \frac{1}{2}\Big),$$

  where $\omega_{\sigma,k}$ is the frequency of the phonon mode k of the crystal in the configuration σ, and $\nu_k = 0, 1, 2, \ldots$ is the occupation number of the mode.

- The electronic excitation energy $E^{\mathrm{el}}_{\sigma\nu}$ is the energy of the electrons above the ground state, and vanishes in the electronic ground state.

This decomposition is the Born–Oppenheimer picture of a crystal. The derivation of Section 4 uses only the labels (σ, ν) and the energies $U_{\sigma\nu}$, so the result does not depend on the harmonic approximation.

The two assumptions of the model. The model rests on two assumptions, and Section 4 gives the mathematical form of each.

- **Assumption 1, separation into configurations.** Every microstate belongs to exactly one configuration σ. This assumption holds when the energy barrier for a defect to hop to another site, or to change the internal state, is large compared with $k_B T$. The crystal then vibrates many times within one configuration before the configuration changes.
- **Assumption 2, non-interacting defects.** The defects are far apart, so each defect changes the crystal by the same amount, whatever the positions of the other defects.

These two statements are assumptions about the crystal and cannot be derived. All the results of Sections 4 and 5 are exact consequences of the two assumptions. Sections 6 and 7 add approximations, each with an error bound.

3 The Grand Canonical Ensemble of the Model

Distribution from the entropy maximization. The crystal exchanges energy, atoms and electrons with the reservoirs, so the constrained averages are the energy, the number of atoms of each species and the number of electrons. The maximum of the Gibbs–Shannon entropy for these constraints is the grand canonical distribution of Section 5 of the note. The distribution has one term $\mu_i N_i$ for each species, as in Section 5.5 of the note, and the term $E_F N_e$ for the electrons,

$$\boxed{p_{\sigma\nu} = \frac{1}{\Xi}\, e^{-\beta\left(U_{\sigma\nu} - \sum_i \mu_i N_{i,\sigma} - E_F N_{e,\sigma}\right)}, \qquad \Xi = \sum_{\sigma}\sum_{\nu} e^{-\beta\left(U_{\sigma\nu} - \sum_i \mu_i N_{i,\sigma} - E_F N_{e,\sigma}\right)},} \tag{3.1}$$

where $p_{\sigma\nu}$ is the probability of the microstate (σ, ν), Ξ is the grand canonical partition function and $\beta = 1/(k_B T)$.

Grand potential and conjugate variables. The thermodynamic potential of the ensemble is the grand potential of Section 5.2 of the note,

$$\boxed{\Phi(T, V, \{\mu_i\}, E_F) = -k_B T \ln \Xi = U - TS - \sum_i \mu_i \langle N_{i,\sigma} \rangle - E_F \langle N_{e,\sigma} \rangle,} \tag{3.2}$$

where U is the internal energy, S is the entropy, and $\langle N_{i,\sigma} \rangle$ and $\langle N_{e,\sigma} \rangle$ are the average numbers of atoms and electrons in the crystal. The differential of the grand potential is

$$\boxed{d\Phi = -S\, dT - P\, dV - \sum_i \langle N_{i,\sigma} \rangle\, d\mu_i - \langle N_{e,\sigma} \rangle\, dE_F,} \tag{3.3}$$

where P is the pressure. Each natural variable therefore has one conjugate variable, the entropy for T, the pressure for V, the number of atoms of species i for μ_i and the number of electrons for E_F. The task is to evaluate Ξ for the model and to read off the four conjugate variables.

4 Evaluation of the Partition Function

Order of the sums. The partition function of Eq. (3.1) is a double sum. The inner sum runs over the states ν of one configuration, and the outer sum runs over the configurations σ. Figure 2 shows this grouping, with each well as one configuration and each ladder as the states ν of that configuration.

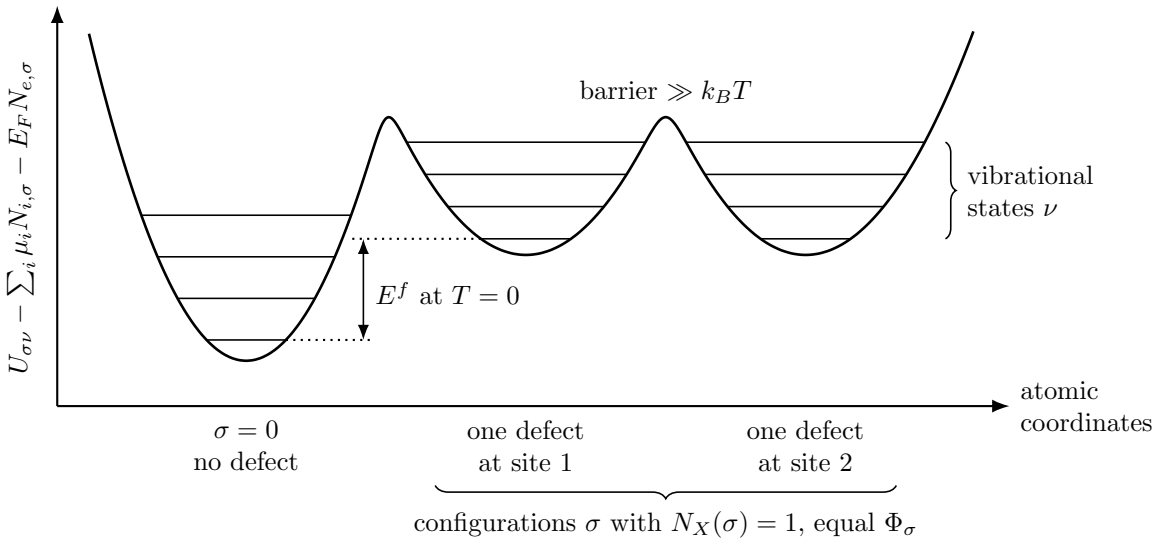


Figure 2: Grouping of the microstates into configurations. Each well: one defect configuration σ. Horizontal lines: vibrational states ν of the crystal held in that configuration. Offset between the lowest levels: formation energy at $T = 0$.

Step 1: the sum over the states ν of one configuration. The particle numbers do not depend on ν, so the reservoir terms come out of the inner sum,

$$\sum_{\nu} e^{-\beta\left(U_{\sigma\nu}-\sum_i \mu_i N_{i,\sigma}-E_F N_{e,\sigma}\right)} = e^{\beta\left(\sum_i \mu_i N_{i,\sigma}+E_F N_{e,\sigma}\right)} \sum_{\nu} e^{-\beta U_{\sigma\nu}}.$$

The remaining sum adds the Boltzmann factors of all the states of the crystal held in the configuration σ, at fixed particle numbers and fixed volume. This sum is the canonical partition function of Section 3 of the note. Here, Z_σ and F_σ denote the canonical partition function and the Helmholtz free energy of the crystal held in the configuration σ,

$$Z_\sigma = \sum_{\nu} e^{-\beta U_{\sigma\nu}}, \qquad F_\sigma = -k_B T \ln Z_\sigma.$$

The free energy F_σ contains an entropy, because the sum counts many states. With the probability of the state ν within the configuration,

$$p_{\nu|\sigma} = \frac{e^{-\beta U_{\sigma\nu}}}{Z_\sigma},$$

the mean energy U_σ and the entropy S_σ of the crystal held in the configuration σ are

$$U_\sigma = \sum_{\nu} p_{\nu|\sigma}\, U_{\sigma\nu},$$

$$S_\sigma = -k_B \sum_{\nu} p_{\nu|\sigma} \ln p_{\nu|\sigma} = k_B \sum_{\nu} p_{\nu|\sigma} \left(\beta U_{\sigma\nu} + \ln Z_\sigma\right) = \frac{U_\sigma}{T} + k_B \ln Z_\sigma,$$

so that

$$F_\sigma = -k_B T \ln Z_\sigma = U_\sigma - T S_\sigma.$$

Here, Φ_σ denotes the grand potential of the crystal held in the configuration σ,

$$\Phi_\sigma = F_\sigma - \sum_i \mu_i N_{i,\sigma} - E_F N_{e,\sigma}.$$

With these definitions, the partition function becomes

$$\boxed{\Xi = \sum_{\sigma} e^{-\beta \Phi_\sigma}.} \tag{4.1}$$

Step 1 is exact under Assumption 1, which is the statement that the labels (σ, ν) cover every microstate exactly once. The grand potentials Φ_{bulk} and $\Phi[X^q]$ of Section 1 of the defect notes are Φ_σ for the perfect configuration and for a configuration with one defect.

The free energy of one configuration for harmonic vibrations. For harmonic vibrations with the electrons in the ground state, the sum over ν runs over every occupation number ν_k independently, and each mode contributes a geometric series,

$$\begin{aligned} Z_\sigma &= e^{-\beta E_\sigma^{\mathrm{stat}}} \prod_k \sum_{\nu_k=0}^{\infty} e^{-\beta\hbar\omega_{\sigma,k}\left(\nu_k+\frac{1}{2}\right)} \\ &= e^{-\beta E_\sigma^{\mathrm{stat}}} \prod_k \frac{e^{-\beta\hbar\omega_{\sigma,k}/2}}{1-e^{-\beta\hbar\omega_{\sigma,k}}} = e^{-\beta E_\sigma^{\mathrm{stat}}} \prod_k \frac{1}{2\sinh\left(\beta\hbar\omega_{\sigma,k}/2\right)}. \end{aligned}$$

The free energy is therefore

$$\boxed{F_\sigma = E_\sigma^{\mathrm{stat}} + k_B T \sum_k \ln\Big[2\sinh\Big(\frac{\hbar\omega_{\sigma,k}}{2k_BT}\Big)\Big].} \tag{4.2}$$

For $T \to 0$, each logarithm tends to $\hbar\omega_{\sigma,k}/(2k_BT)$, and

$$F_\sigma \to E_\sigma^{\mathrm{stat}} + \sum_k \frac{\hbar\omega_{\sigma,k}}{2},$$

which is the static energy plus the zero-point energy, the result $F = U = E_{\mathrm{tot}} + E_{\mathrm{ZPE}}$ of Section 1 of the defect notes.

Step 2: non-interacting defects and the formation energy. Here, F_{bulk} denotes F_σ of the perfect configuration, $F[X^q]$ denotes F_σ of a configuration with one defect, and ΔF denotes the difference,

$$\Delta F = F[X^q] - F_{\mathrm{bulk}}.$$

All the sites of the lattice are equivalent, and the internal states have equal energies, so ΔF does not depend on the site or on the internal state of the defect. Assumption 2 states that every further defect changes the free energy by the same amount,

$$\boxed{F_\sigma = F_{\mathrm{bulk}} + N_X(\sigma)\,\Delta F.} \tag{4.3}$$

This assumption requires that the static energies of distant defects add, and that each defect changes the phonon frequencies only in the surroundings of the defect. The assumption fails at high concentrations and for charged defects with an unscreened Coulomb interaction. The particle numbers of Eq. (2.1) then give

$$\begin{aligned} \Phi_\sigma &= F_{\mathrm{bulk}} + N_X(\sigma)\,\Delta F - \sum_i \mu_i\big[N_i + n_i\,N_X(\sigma)\big] - E_F\big[N_e - q\,N_X(\sigma)\big] \\ &= \Big[F_{\mathrm{bulk}} - \sum_i \mu_i N_i - E_F N_e\Big] + N_X(\sigma)\Big[\Delta F - \sum_i \mu_i n_i + qE_F\Big]. \end{aligned}$$

The first bracket is the grand potential Φ_{bulk} of the perfect configuration. The second bracket is the formation energy of one defect,

$$\boxed{E^f[X^q] = F[X^q] - F_{\text{bulk}} - \sum_i \mu_i n_i + qE_F = \Phi[X^q] - \Phi_{\text{bulk}},} \tag{4.4}$$

the definition of Section 1 of the defect notes. The formation energy is a function of the natural variables. The temperature and the volume enter through ΔF, and the chemical potentials and the Fermi level enter through the reservoir terms. For brevity, E^f replaces $E^f[X^q]$ until Section 9, because only one type of defect appears. Every configuration therefore has

$$\boxed{\Phi_\sigma = \Phi_{\text{bulk}} + N_X(\sigma)\, E^f.} \tag{4.5}$$

Step 3: the sum over the configurations. Equations (4.1) and (4.5) give

$$\Xi = e^{-\beta\Phi_{\text{bulk}}} \sum_\sigma e^{-\beta E^f N_X(\sigma)}.$$

The number of defects is a sum over the sites, so the exponential is a product over the sites. The sum over σ is a sum over every σ_j independently, and the sum of the products is the product of the sums,

$$\begin{aligned}
\sum_\sigma e^{-\beta E^f N_X(\sigma)} &= \sum_{\sigma_1=0}^{g_X} \cdots \sum_{\sigma_N=0}^{g_X} \prod_{j=1}^{N} e^{-\beta E^f \left(1-\delta_{\sigma_j,0}\right)} \\
&= \prod_{j=1}^{N} \sum_{\sigma_j=0}^{g_X} e^{-\beta E^f \left(1-\delta_{\sigma_j,0}\right)} \\
&= \prod_{j=1}^{N} \left(1 + g_X\, e^{-\beta E^f}\right) = \left(1 + g_X\, e^{-\beta E^f}\right)^N.
\end{aligned}$$

Each site contributes the weight 1 for the perfect state and the weight $e^{-\beta E^f}$ for each of the g_X defect states. Here, ζ_X denotes the total weight of the defect states of one site,

$$\zeta_X = g_X\, e^{-E^f/k_BT}.$$

The partition function and the grand potential of the model are

$$\boxed{\Xi = e^{-\Phi_{\text{bulk}}/k_BT} \left(1+\zeta_X\right)^N, \qquad \Phi = \Phi_{\text{bulk}} - Nk_BT \ln\left(1+\zeta_X\right).} \tag{4.6}$$

The result uses no Stirling approximation and no dilute limit. The crystal behaves as N independent sites, each with one perfect state and g_X defect states.

5 Conjugate Variables

State of one site and the defect concentration. The probability of a configuration is the sum of $p_{\sigma\nu}$ over ν,

$$p_\sigma = \sum_\nu p_{\sigma\nu} = \frac{e^{-\beta\Phi_\sigma}}{\Xi} = \prod_{j=1}^{N} \frac{e^{-\beta E^f\left(1-\delta_{\sigma_j,0}\right)}}{1+\zeta_X}.$$

The probability is a product over the sites, so the sites are statistically independent. A site is perfect with the probability $1/(1+\zeta_X)$ and holds a defect in a given internal state with the probability $e^{-\beta E^f}/(1+\zeta_X)$. The defect concentration is the average number of defects per site,

$$\boxed{c \equiv \frac{\langle N_X\rangle}{N} = \frac{\zeta_X}{1+\zeta_X} = \frac{1}{1+g_X^{-1}\,e^{E^f/k_BT}}.} \tag{5.1}$$

The concentration has the form of the Fermi–Dirac occupation, because a site holds at most one defect, and never exceeds one (Fig. 3b). The same concentration follows from a derivative of the grand potential,

$$\left.\frac{\partial\Phi}{\partial E^f}\right|_T = -Nk_BT\,\frac{1}{1+\zeta_X}\,\frac{\partial\zeta_X}{\partial E^f} = N\,\frac{\zeta_X}{1+\zeta_X} = Nc,$$

so the formation energy and the number of defects are a conjugate pair.

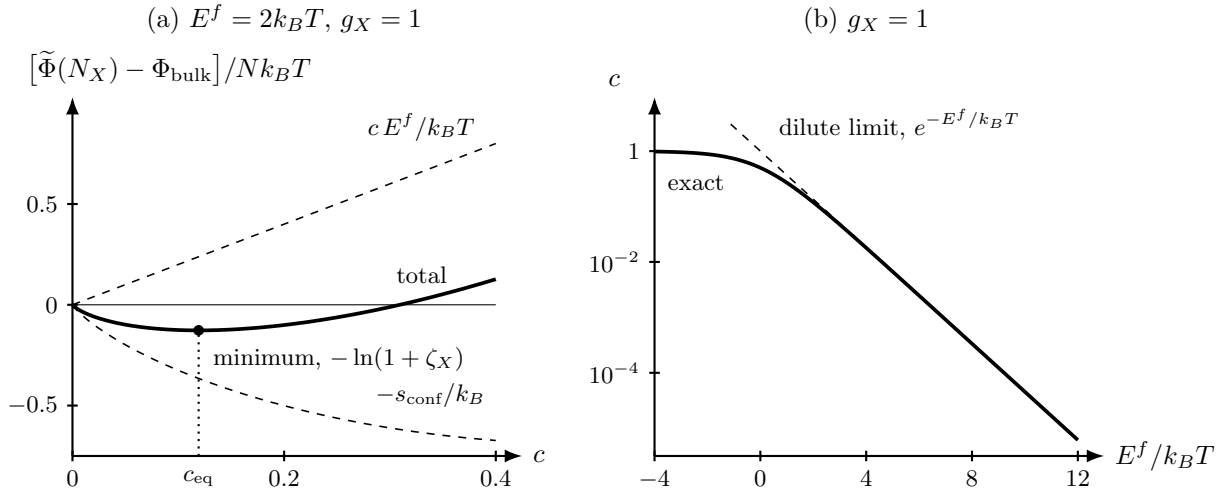


Figure 3: (a) Constrained grand potential per site of Section 7 against the defect concentration for $E^f = 2k_BT$ and $g_X = 1$, with the energy term, the entropy term and the total. Dot: minimum at the equilibrium concentration. (b) Defect concentration against E^f/k_BT for $g_X = 1$. Solid line: exact result. Dashed line: dilute limit.

Differential of the grand potential. Here, $\Phi_{\rm def}$ denotes the defect part of the grand potential,

$$\Phi_{\rm def} = \Phi - \Phi_{\rm bulk} = -Nk_BT \ln\big(1+\zeta_X\big).$$

The defect part depends on the natural variables through E^f and through the explicit temperature, so

$$d\Phi = d\Phi_{\rm bulk} + \left.\frac{\partial\Phi_{\rm def}}{\partial T}\right|_{E^f} dT + Nc\, dE^f.$$

The explicit temperature derivative is

$$\begin{aligned}\left.\frac{\partial\Phi_{\rm def}}{\partial T}\right|_{E^f} &= -Nk_B \ln\big(1+\zeta_X\big) - Nk_BT\,\frac{1}{1+\zeta_X}\,\frac{\zeta_X E^f}{k_BT^2}\\ &= -Nk_B \ln\big(1+\zeta_X\big) - Nc\,\frac{E^f}{T}.\end{aligned}$$

Equation (5.1) gives the two relations

$$\boxed{\ln\big(1+\zeta_X\big) = -\ln(1-c), \qquad \frac{E^f}{k_BT} = \ln g_X - \ln c + \ln(1-c),} \tag{5.2}$$

and the derivative becomes

$$\begin{aligned}\left.\frac{\partial\Phi_{\rm def}}{\partial T}\right|_{E^f} &= -Nk_B\big[-\ln(1-c) + c\ln g_X - c\ln c + c\ln(1-c)\big]\\ &= -Nk_B\big[c\ln g_X - c\ln c - (1-c)\ln(1-c)\big] = -N\, s_{\rm conf}(c),\end{aligned}$$

with the configurational entropy per site

$$\boxed{s_{\rm conf}(c) = k_B\big[c\ln g_X - c\ln c - (1-c)\ln(1-c)\big].} \tag{5.3}$$

The formation energy of Eq. (4.4) has the differential

$$dE^f = -S^f\, dT - \frac{B_T V^f}{V}\, dV - \sum_i n_i\, d\mu_i + q\, dE_F,$$

which defines the formation entropy S^f and the formation volume V^f of one defect,

$$S^f = -\left.\frac{\partial E^f}{\partial T}\right|_V = -\left.\frac{\partial \Delta F}{\partial T}\right|_V, \qquad V^f = -\frac{V}{B_T}\left.\frac{\partial E^f}{\partial V}\right|_T, \qquad B_T = -V\left.\frac{\partial P_{\rm bulk}}{\partial V}\right|_T,$$

where B_T is the isothermal bulk modulus of the perfect crystal. The formation entropy is the change of the vibrational and electronic entropy S_σ when one defect forms. The perfect configuration has fixed particle numbers, so

$$d\Phi_{\rm bulk} = -S_{\rm bulk}\, dT - P_{\rm bulk}\, dV - \sum_i N_i\, d\mu_i - N_e\, dE_F,$$

where $S_{\rm bulk}$ and $P_{\rm bulk}$ are the entropy and the pressure of the perfect crystal. The sum of the three contributions is

$$\boxed{\begin{aligned} d\Phi = & -\left[S_{\rm bulk} + N\, s_{\rm conf}(c) + Nc\, S^f\right] dT - \left[P_{\rm bulk} + Nc\, \frac{B_T V^f}{V}\right] dV \\ & - \sum_i \left[N_i + n_i\, Nc\right] d\mu_i - \left[N_e - q\, Nc\right] dE_F. \end{aligned}} \tag{5.4}$$

The comparison with Eq. (3.3) gives all the conjugate variables, which Table 1 lists.

Table 1: Conjugate variables of the grand potential of the lattice model.

Natural variable	Conjugate variable	Perfect crystal	Defects
T	$S = -\dfrac{\partial \Phi}{\partial T}$	$S_{\rm bulk}$	$N\, s_{\rm conf}(c) + Nc\, S^f$
V	$P = -\dfrac{\partial \Phi}{\partial V}$	$P_{\rm bulk}$	$Nc\, \dfrac{B_T V^f}{V}$
μ_i	$\langle N_{i,\sigma} \rangle = -\dfrac{\partial \Phi}{\partial \mu_i}$	N_i	$n_i\, Nc$
E_F	$\langle N_{e,\sigma} \rangle = -\dfrac{\partial \Phi}{\partial E_F}$	N_e	$-q\, Nc$

Particle numbers. The crystal holds the atoms and electrons of the perfect crystal, plus n_i atoms and minus q electrons for each of the Nc defects. The derivatives of Φ therefore agree with the direct averages of Eq. (2.1),

$$\boxed{\langle N_{i,\sigma} \rangle = N_i + n_i\, Nc, \qquad \langle N_{e,\sigma} \rangle = N_e - q\, Nc.} \tag{5.5}$$

Entropy and the origin of the configurational entropy. The entropy has three contributions,

$$\boxed{S = S_{\rm bulk} + Nc\, S^f + N\, s_{\rm conf}(c).} \tag{5.6}$$

The first two contributions are vibrational and electronic. The third contribution is the entropy of the distribution over the configurations. The probability of a microstate is the product $p_{\sigma\nu} =$

$p_\sigma\, p_{\nu|\sigma}$, so the Gibbs–Shannon entropy splits into two parts,

$$\begin{aligned} S &= -k_B \sum_\sigma \sum_\nu p_\sigma\, p_{\nu|\sigma} \big[\ln p_\sigma + \ln p_{\nu|\sigma} \big] \\ &= -k_B \sum_\sigma p_\sigma \ln p_\sigma + \sum_\sigma p_\sigma \Big[- k_B \sum_\nu p_{\nu|\sigma} \ln p_{\nu|\sigma} \Big] \\ &= -k_B \sum_\sigma p_\sigma \ln p_\sigma + \sum_\sigma p_\sigma\, S_\sigma . \end{aligned}$$

The second part is the average of the entropy S_σ of Step 1. With Eq. (4.3), $S_\sigma = S_{\text{bulk}} + N_X(\sigma)\, S^f$, and the average is $S_{\text{bulk}} + Nc\, S^f$. The first part is the entropy of the configurations. The sites are independent, so this entropy is N times the entropy of one site (Section 3.2 of the note). One site is perfect with the probability $1 - c$ and is in each of the g_X defect states with the probability c/g_X, so

$$\begin{aligned} -k_B \sum_\sigma p_\sigma \ln p_\sigma &= -N k_B \Big[(1-c) \ln(1-c) + g_X\, \frac{c}{g_X} \ln \frac{c}{g_X} \Big] \\ &= N k_B \big[c \ln g_X - c \ln c - (1-c) \ln(1-c) \big] = N\, s_{\text{conf}}(c). \end{aligned}$$

The direct evaluation therefore reproduces Eq. (5.6). The configurational entropy is not an additional postulate. This entropy is the part of the Gibbs–Shannon entropy that belongs to the distribution over the configurations, and the vibrational entropy is the part that belongs to the states within one configuration.

Internal energy. The internal energy follows from Eq. (3.2),

$$U = \Phi + TS + \sum_i \mu_i \langle N_{i,\sigma} \rangle + E_F \langle N_{e,\sigma} \rangle .$$

The terms of the perfect crystal add up to the internal energy U_{bulk} of the perfect crystal. For the defect terms, Eq. (5.2) gives

$$-k_B T \ln \big(1 + \zeta_X \big) + T\, s_{\text{conf}}(c) = k_B T\, c \, \big[\ln g_X - \ln c + \ln(1-c) \big] = c\, E^f ,$$

so that

$$\begin{aligned} U &= U_{\text{bulk}} + Nc \Big[E^f + TS^f + \sum_i \mu_i n_i - q E_F \Big] \\ &= U_{\text{bulk}} + Nc \, \big[\Delta F + TS^f \big] . \end{aligned}$$

Here, $\Delta U = \Delta F + TS^f$ denotes the change of the internal energy when one defect forms, which contains the static, the vibrational and the electronic energy. The internal energy is

$$\boxed{U = U_{\text{bulk}} + Nc\,\Delta U.} \tag{5.7}$$

The internal energy contains no reservoir term and no configurational term, because the arrangement of the defects on the sites costs no energy.

Pressure and the meaning of the formation volume. The pressure of the crystal is

$$\boxed{P = P_{\text{bulk}} + Nc\,\frac{B_T V^f}{V}.} \tag{5.8}$$

At fixed volume, each defect raises the pressure by $B_T V^f/V$. If the crystal is released to the original pressure P_{bulk}, the volume changes by ΔV, with

$$-\frac{B_T}{V}\,\Delta V + Nc\,\frac{B_T V^f}{V} = 0 \qquad \Longrightarrow \qquad \Delta V = Nc\,V^f$$

to first order in c. The formation volume V^f is therefore the volume change per defect at fixed pressure, with the exchanged atoms in the reservoirs. For a vacancy, V^f does not include the volume of the removed atom, because this atom is in the reservoir.

Fluctuation of the number of defects. The number of defects is a sum of N independent site variables, each equal to one with the probability c and to zero with the probability $1-c$. The variance of one site variable is $c(1-c)$, and the variances of independent variables add, so

$$\boxed{\text{Var}(N_X) = Nc\,(1-c), \qquad \frac{\sqrt{\text{Var}(N_X)}}{\langle N_X\rangle} = \sqrt{\frac{1-c}{Nc}}.} \tag{5.9}$$

The same variance follows from the fluctuation formula of Section 5.1 of the note, with $-E^f$ in the role of the chemical potential,

$$\text{Var}(N_X) = -k_B T\,\frac{\partial\langle N_X\rangle}{\partial E^f}\bigg|_T = -k_B T\,N\,\frac{\partial c}{\partial E^f}\bigg|_T = Nc\,(1-c).$$

For $N = 10^{22}$ sites and $c = 10^{-6}$, the relative fluctuation is 10^{-8}, so the number of defects is sharply defined.

6 Approximations with Error Bounds

The dilute limit. For $\zeta_X \geq 0$, the product $(1-\zeta_X)(1+\zeta_X) = 1-\zeta_X^2$ does not exceed one, so

$$1 - \zeta_X \;\leq\; \frac{1}{1+\zeta_X} \;\leq\; 1.$$

Multiplication by ζ_X bounds the concentration of Eq. (5.1), and integration of the three terms from 0 to ζ_X bounds the logarithm of Eq. (4.6),

$$\boxed{\zeta_X - \zeta_X^2 \;\leq\; c \;\leq\; \zeta_X, \qquad \zeta_X - \frac{\zeta_X^2}{2} \;\leq\; \ln\bigl(1+\zeta_X\bigr) \;\leq\; \zeta_X.} \tag{6.1}$$

The dilute limit keeps the first order in ζ_X,

$$\boxed{c \approx g_X\, e^{-E^f/k_BT}, \qquad \Phi \approx \Phi_{\text{bulk}} - Nk_BT\, c,} \tag{6.2}$$

with a relative error of at most ζ_X in each expression. These results are Eqs. (41) and (40) of Freysoldt *et al.* [1]. For $E^f = 1$ eV, $T = 1000$ K and $g_X = 1$, the exponent is $E^f/k_BT = 11.6$ and $\zeta_X = 9\times10^{-6}$, so the dilute limit is accurate to one part in 10^5. The configurational entropy has the dilute form

$$\boxed{s_{\text{conf}}(c) \approx k_B\bigl[c\ln g_X - c\ln c + c\bigr],} \tag{6.3}$$

which is Eq. (7) of the review. The error is the function

$$h(c) = c + (1-c)\ln(1-c), \qquad h'(c) = -\ln(1-c) \geq 0, \qquad h''(c) = \frac{1}{1-c} \leq 2 \quad \text{for } c \leq \tfrac{1}{2},$$

with $h(0) = h'(0) = 0$, so that $0 \leq h(c) \leq c^2$ for $c \leq 1/2$. The dilute form therefore overestimates s_{conf} by at most k_Bc^2.

The formation energy at $T = 0$ in the concentration. A DFT calculation gives the formation energy at $T = 0$. At fixed volume, chemical potentials and Fermi level, the definition of the formation entropy gives the exact relation

$$E^f(T) = E^f(0) - \int_0^T S^f(T')\,dT'.$$

Here, $\bar{S}^f$ denotes the formation entropy averaged over the temperature,

$$\bar{S}^f(T) = \frac{1}{T}\int_0^T S^f(T')\,dT'.$$

The dilute concentration of Eq. (6.2) becomes

$$\boxed{c \approx g_X \, e^{\bar{S}^f(T)/k_B} \, e^{-E^f(0)/k_B T}.} \tag{6.4}$$

The formation energy at $T = 0$ gives the last factor. The factor $e^{\bar{S}^f/k_B}$ is the exact price of the neglected temperature dependence. Formation entropies of a few k_B are typical, and the value $2.4\,k_B$ quoted in the review for the vacancy in aluminum corresponds to a factor of $e^{2.4} = 11$. If a reservoir is a phase with a chemical potential that depends on the temperature, E^f changes along the path by

$$\frac{dE^f}{dT} = -S^f - \sum_i n_i \frac{d\mu_i}{dT},$$

and Eq. (6.4) holds with $-dE^f/dT$ in place of S^f.

7 The Minimization Route of the Review

The constrained grand potential. Freysoldt *et al.* [1] obtained the concentration from the minimum of a potential at a fixed number of defects. This potential follows from the sum of Eq. (4.1), with the configurations grouped by the number of defects,

$$\Xi = \sum_{N_X=0}^{N} \sum_{\sigma:\, N_X(\sigma)=N_X} e^{-\beta\Phi_\sigma} = \sum_{N_X=0}^{N} W(N_X)\, e^{-\beta\left(\Phi_{\text{bulk}}+N_X E^f\right)},$$

where $W(N_X)$ is the number of configurations with N_X defects. A configuration with N_X defects is fixed by the choice of N_X sites out of N and of one internal state for each defect, so

$$\boxed{W(N_X) = g_X^{N_X} \, \frac{N!}{N_X!\,(N-N_X)!}.} \tag{7.1}$$

All these configurations have the same Φ_σ and are equally probable. The entropy of W equally probable configurations is the microcanonical entropy of Section 2 of the note,

$$S_{\text{conf}}(N_X) = k_B \ln W(N_X).$$

Here, $\widetilde{\Phi}(N_X)$ denotes the constrained grand potential, the grand potential of the ensemble restricted to N_X defects,

$$\boxed{\widetilde{\Phi}(N_X) = -k_B T \ln\left[W(N_X)\, e^{-\beta\left(\Phi_{\text{bulk}}+N_X E^f\right)}\right] = \Phi_{\text{bulk}} + N_X\, E^f - T\, S_{\text{conf}}(N_X).} \tag{7.2}$$

The potential F^{conf} of Eq. (8) of the review is $\widetilde{\Phi}(N_X) - \Phi_{\text{bulk}}$ per site. The term $N_X E^f$ contains the reservoir terms, so this potential is a constrained grand potential and not a Helmholtz free energy.

The largest term of the sum. The partition function is a sum of $N+1$ positive terms,

$$\Xi = \sum_{N_X=0}^{N} e^{-\beta \widetilde{\Phi}(N_X)}.$$

A sum of $N+1$ positive terms is at least the largest term and at most $N+1$ times the largest term. The largest term belongs to the smallest $\widetilde{\Phi}(N_X)$, so the logarithm of the two bounds gives

$$\boxed{\min_{N_X} \widetilde{\Phi}(N_X) - k_B T \ln(N+1) \;\le\; \Phi \;\le\; \min_{N_X} \widetilde{\Phi}(N_X).} \tag{7.3}$$

The grand potential per site equals the minimum of the constrained potential per site up to $k_B T \ln(N+1)/N$, which is $5 \times 10^{-21}\, k_B T$ for $N = 10^{22}$. The minimization of the review is therefore exact in the thermodynamic limit. The sharp distribution of Eq. (5.9) expresses the same fact.

The Stirling approximation with a bound. The logarithm of a factorial is a sum of logarithms. The logarithm is an increasing function, so each term lies between two integrals,

$$\int_{k-1}^{k} \ln t \, dt \;\le\; \ln k \;\le\; \int_{k}^{k+1} \ln t \, dt.$$

The sum of the lower bounds from $k=1$ to n, and the sum of the upper bounds from $k=1$ to $n-1$ plus the last term $\ln n$, give

$$\int_{0}^{n} \ln t \, dt \;\le\; \ln n! \;\le\; \int_{1}^{n} \ln t \, dt + \ln n.$$

With the integral $\int \ln t \, dt = t \ln t - t$, the bounds are

$$\boxed{n \ln n - n \;\le\; \ln n! \;\le\; n \ln n - n + 1 + \ln n.} \tag{7.4}$$

The Stirling formula $\ln n! \approx n \ln n - n$ therefore has an error between 0 and $1 + \ln n$. For $0 < N_X < N$ and $c = N_X/N$, the leading terms of the three factorials in $W(N_X)$ are

$$\begin{aligned} \ln W(N_X) &\approx N_X \ln g_X + N \ln N - N_X \ln N_X - (N - N_X) \ln(N - N_X) \\ &= N\big[c \ln g_X - c \ln c - (1-c) \ln(1-c)\big], \end{aligned}$$

with a total error of at most $2(1+\ln N)$. The terms $-N$, $+N_X$ and $+(N-N_X)$ cancel. The constrained potential per site is therefore

$$\boxed{\frac{\widetilde{\Phi}(N_X)-\Phi_{\text{bulk}}}{N} = c\,E^f - T\,s_{\text{conf}}(c) + \mathcal{O}\Big(\frac{k_BT\ln N}{N}\Big),} \tag{7.5}$$

with the same $s_{\text{conf}}(c)$ as in Eq. (5.3).

The minimum reproduces the exact result. The derivative of the leading term of Eq. (7.5) is

$$\frac{d}{dc}\big[c\,E^f - T\,s_{\text{conf}}(c)\big] = E^f - k_BT\big[\ln g_X - \ln c + \ln(1-c)\big],$$

and the second derivative, $k_BT\,[1/c+1/(1-c)]$, is positive, so the stationary point is a minimum (Fig. 3a). The derivative vanishes for

$$\frac{c}{1-c} = g_X\,e^{-E^f/k_BT} = \zeta_X \qquad\Longrightarrow\qquad c = \frac{\zeta_X}{1+\zeta_X},$$

which is the concentration of Eq. (5.1). By Eq. (5.2), the value at the minimum is

$$c\,E^f - T\,s_{\text{conf}}(c) = -k_BT\ln\big(1+\zeta_X\big),$$

which is the defect part of Eq. (4.6) per site. The minimization route therefore agrees with the direct sum. The review used the dilute entropy of Eq. (6.3) in the minimization, which gives the dilute results of Eq. (6.2).

8 Fixed Pressure

The potential at fixed pressure. Experiments fix the pressure and not the volume. The reservoirs fix the chemical potentials and the Fermi level in both cases. The potential with the natural variables $(T, P, \{\mu_i\}, E_F)$ is the Legendre transform of Φ in the volume (Section 4.4 of the note). Here, Φ_P denotes this potential,

$$\Phi_P(T,P,\{\mu_i\},E_F) = \Phi + PV, \qquad P = -\left.\frac{\partial\Phi}{\partial V}\right|_{T,\{\mu_i\},E_F}.$$

For a fluid, this potential vanishes identically, because no extensive variable would fix the size of the system (Section 5.7 of the note). In the lattice model, the fixed number of sites N fixes the size, so Φ_P is a proper potential. The Gibbs energy G of the review is this potential, because the review subtracted the reservoir terms.

The formation energy at fixed pressure. The perfect crystal at the pressure P has the volume V, with

$$\left.\frac{\partial \Phi_{\text{bulk}}}{\partial V}\right|_V = -P.$$

A crystal with one defect has the grand potential $\Phi_{\text{bulk}}(V') + E^f(V')$ at the volume V' and relaxes to the volume that minimizes the potential at fixed pressure. Here, E_P^f denotes the formation energy at fixed pressure,

$$E_P^f = \min_{V'} \Big[\Phi_{\text{bulk}}(V') + E^f(V') + PV'\Big] - \Big[\Phi_{\text{bulk}}(V) + PV\Big].$$

With Eq. (4.4) written out, this definition reads

$$E_P^f = F[X^q](V') - F_{\text{bulk}}(V) + P\,(V' - V) - \sum_i \mu_i n_i + qE_F,$$

which is the Gibbs energy of formation G^f of Eq. (23) of the review. The expansion of the two terms about the volume V to second order in $V' - V$ uses the definitions of B_T and V^f of Section 5,

$$\Phi_{\text{bulk}}(V') + PV' \approx \Phi_{\text{bulk}}(V) + PV + \frac{B_T}{2V}\,(V' - V)^2,$$
$$E^f(V') \approx E^f(V) - \frac{B_T V^f}{V}\,(V' - V).$$

The sum is a minimum for $V' - V = V^f$, and the value at the minimum gives

$$\boxed{E_P^f = E^f(V) - \frac{B_T\,(V^f)^2}{2V}.} \tag{8.1}$$

The difference between the two formation energies is proportional to $1/V$ and vanishes for one defect in a large crystal. For $B_T = 100$ GPa, $V^f = 10^{-29}$ m^3 and a volume of 2×10^{-29} m^3 per site, the difference is 1.6 eV divided by the number of sites, which is 16 meV for a supercell of 100 sites and negligible for a macroscopic crystal. The term $P\,(V' - V)$ does not appear in Eq. (8.1), because the change of $F[X^q]$ between the volumes V and V' cancels this term to first order.

Finite concentration at fixed pressure. For N_X defects, the same expansion gives the relaxation $V' - V = N_X V^f$ and a correction of $c\,B_T (V^f)^2/(2v)$ per defect, where $v = V/N$ is the volume per site. This correction is an interaction between the defects through the strain of the lattice. The correction is of first order in c and is neglected under Assumption 2. All the results of Sections 4 to 7 therefore hold at fixed pressure, with E^f evaluated at the volume of the perfect crystal at the pressure P and the temperature T.

9 Charge States and the Fermi Level

Several charge states on the same sites. A defect X can take several charge states q. From here on, $E^f[X^q]$ replaces E^f, and g_q and ζ_q replace g_X and ζ_X for the charge state q. Each site is then perfect or holds the defect in one charge state q and one of the g_q internal states. The derivation of Section 4 goes through unchanged, and each site contributes the factor $1 + \sum_q \zeta_q$, with

$$\zeta_q = g_q \, e^{-E^f[X^q]/k_BT}.$$

The grand potential and the concentration c_q of the charge state q are

$$\boxed{\Phi = \Phi_{\rm bulk} - N k_B T \ln\Big(1 + \sum_q \zeta_q\Big), \qquad c_q = \frac{\zeta_q}{1 + \sum_{q'} \zeta_{q'}}.} \tag{9.1}$$

The ratio of the concentrations of two charge states is

$$\boxed{\frac{c_q}{c_{q'}} = \frac{g_q}{g_{q'}} \, e^{-\left(E^f[X^q] - E^f[X^{q'}]\right)/k_BT} = \frac{g_q}{g_{q'}} \, e^{-(q-q')[E_F - \varepsilon(q/q')]/k_BT},} \tag{9.2}$$

where $\varepsilon(q/q')$ is the charge-state transition level of Section 2 of the defect notes, evaluated with the formation energies at the temperature T. For equal numbers of internal states, the two charge states have equal concentrations when the Fermi level is at the transition level. The change of the dominant charge state at the transition level is spread over a few k_BT in the Fermi level.

The condition that fixes the Fermi level. The Fermi level is a natural variable of Φ, set by the electron reservoir. A real crystal has no external electron reservoir. The electrons released by the defects enter the bands or other defects, and the crystal stays neutral. Here, $\mathcal{N}_e$ and $\mathcal{N}_h$ denote the numbers of free electrons and free holes in the bands of the crystal region. The neutrality condition is

$$\boxed{\sum_q q \, N c_q + \mathcal{N}_h - \mathcal{N}_e = 0,} \tag{9.3}$$

which is Eq. (36) of the review. The concentrations c_q and the carrier numbers depend on E_F, so this condition determines E_F. The free carriers add a factor to Ξ that does not depend on the defect configuration, as long as the carriers and the defects do not interact, so the defect factor of Eq. (4.6) is unchanged.

10 Dictionary for the Review

Translation of the symbols and equations. Table 2 relates the symbols and equations of Freysoldt *et al.* [1] to the results of this note.

Table 2: Symbols and equations of the review and the corresponding results of this note.

Review	This note	Remark
E^f, Eq. (1), with E_{tot}	E^f at $T = 0$, Eq. (4.4) with $F_\sigma \to E_\sigma^{\text{stat}}$	The term E_{corr} of the review corrects the finite supercell of the DFT calculation and is absent in the model.
n, N, g	N_X, N, g_X	The review counted g positions per lattice site.
W, Eq. (4)	$W(N_X)$	Different exact counts, same dilute limit $(g_X N)^{N_X}/N_X!$.
$S^{\text{conf}}(c)$, Eq. (7)	Eq. (6.3)	Dilute limit of Eq. (5.3).
$F^{\text{conf}}(c)$, Eq. (8)	$[\widetilde{\Phi}(N_X) - \Phi_{\text{bulk}}]/N$, Eq. (7.5)	Constrained grand potential per site, with E^f at $T = 0$.
G^f, Eq. (23)	E_P^f, Eq. (8.1)	Equal to E^f at the same state up to $B_T(V^f)^2/2V$.
H^f, S^f, Eq. (42)	$E^f + TS^f$, S^f	With the temperature dependence of the reservoirs included.
G, Eq. (39)	$\widetilde{\Phi}(N_X)/N$ at fixed pressure	Contains the reservoir terms.
$G = G[\text{bulk}] - k_B T c^{\text{eq}}$, Eq. (40)	Φ per site, Eq. (6.2)	Dilute limit of Eq. (4.6).
c^{eq}, Eq. (41)	c, Eq. (6.2)	Dilute limit of Eq. (5.1).

The counting of the review. Equation (4) of the review counted the arrangements of N_X defects on gN positions, with g positions for each lattice site. This counting applies when the positions of one lattice site can be occupied independently, as for interstitial positions. The lattice model covers this case with gN sites in place of N and one internal state in place of g_X. The concentration per position is then $1/(1 + e^{E^f/k_B T})$, and the concentration per lattice site in the dilute limit is $g\, e^{-E^f/k_B T}$, the same as Eq. (6.2).

List of symbols. Table 3 lists the symbols of this note.

Table 3: Symbols of this note.

Symbol	Meaning
N, V, v	Number of lattice sites, volume, volume per site V/N
T, β, P	Temperature, $1/(k_B T)$, pressure
μ_i, E_F	Chemical potential of the reservoir of species i, Fermi level of the electron reservoir
X^q, g_X	Defect X in the charge state q, number of internal states of one defect
n_i, q	Atoms of species i added to the crystal and electrons released to the reservoir when one defect forms
σ_j, σ, $N_X(\sigma)$	State of site j, defect configuration, number of defects in the configuration
ν, ν_k, $\omega_{\sigma,k}$	State of the crystal held in one configuration, occupation number and frequency of the phonon mode k
$U_{\sigma\nu}$	Energy of the microstate (σ, ν), the sum of E_σ^{stat}, $E_{\sigma\nu}^{\text{vib}}$ and $E_{\sigma\nu}^{\text{el}}$
N_i, N_e, $N_{i,\sigma}$, $N_{e,\sigma}$	Numbers of atoms and electrons in the perfect crystal and in the configuration σ
$p_{\sigma\nu}$, p_σ, $p_{\nu\|\sigma}$	Probability of a microstate, of a configuration, and of a state within a configuration
Ξ, Z_σ	Grand canonical partition function, canonical partition function of one configuration
F_σ, U_σ, S_σ, Φ_σ	Helmholtz free energy, energy, entropy and grand potential of the crystal held in the configuration σ
F_{bulk}, Φ_{bulk}, S_{bulk}, P_{bulk}, U_{bulk}	Quantities of the perfect crystal
ΔF, ΔU	Change of the free energy and of the internal energy when one defect forms
E^f, E_P^f	Formation energy at fixed volume and at fixed pressure
S^f, $\bar{S}^f$, V^f, B_T	Formation entropy, the temperature average of S^f, formation volume, isothermal bulk modulus
ζ_X, c	Weight of the defect states of one site, defect concentration
Φ, Φ_{def}, Φ_P	Grand potential, the defect part of Φ, grand potential at fixed pressure
S, s_{conf}, S_{conf}	Entropy, configurational entropy per site and of the crystal
$W(N_X)$, $\widetilde{\Phi}(N_X)$	Number of configurations with N_X defects, constrained grand potential

References

[1] C. Freysoldt, B. Grabowski, T. Hickel, J. Neugebauer, G. Kresse, A. Janotti and C. G. Van de Walle, "First-principles calculations for point defects in solids," Rev. Mod. Phys. **86**, 253 (2014).